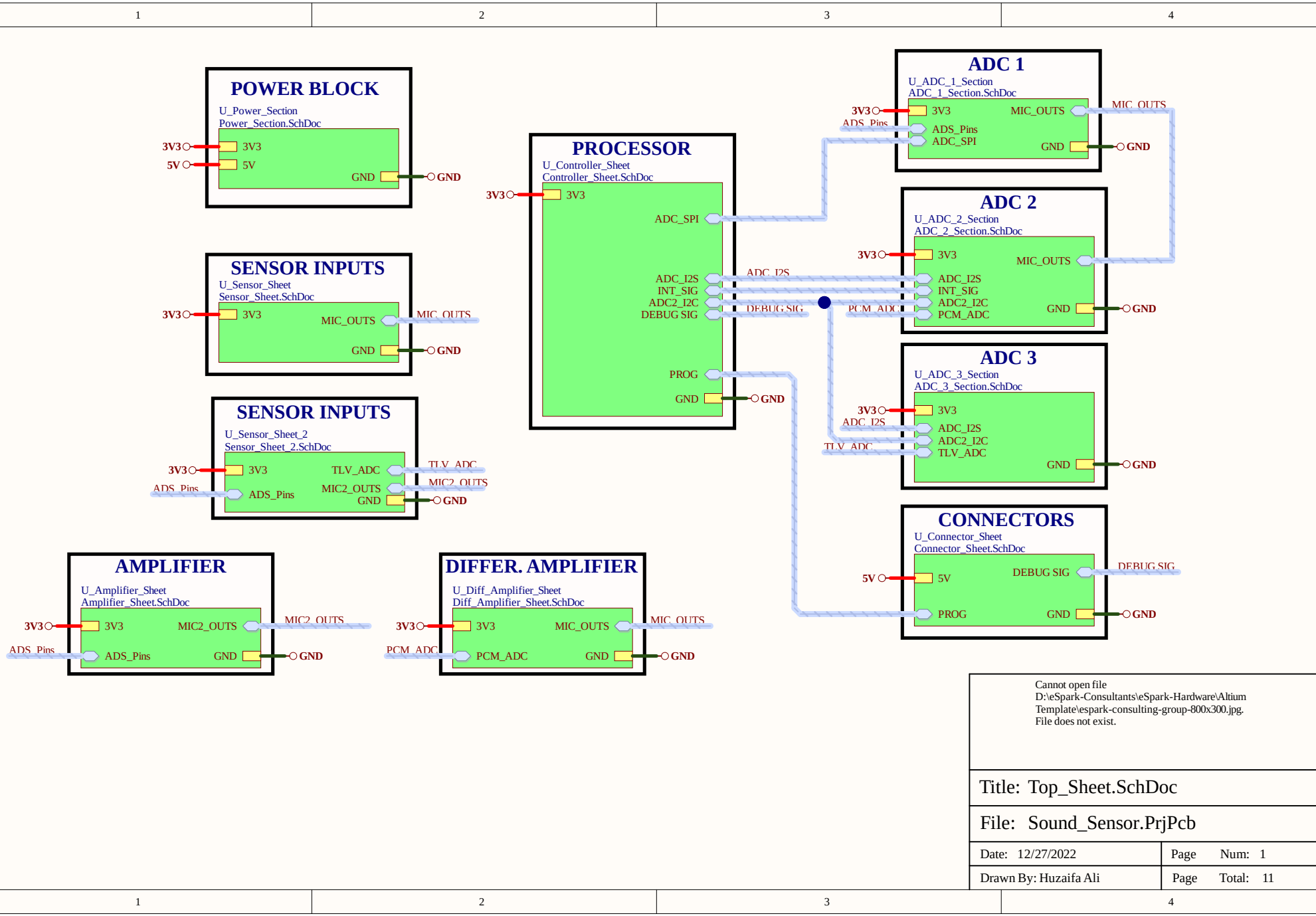
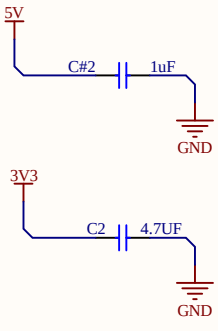


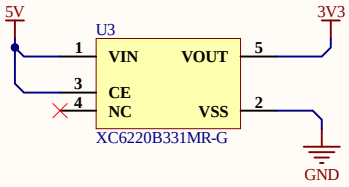
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Title: Top_Sheet.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 12/27/2022	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 11

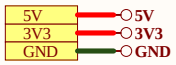
DECOUPLING CAPS



3.3V LDO



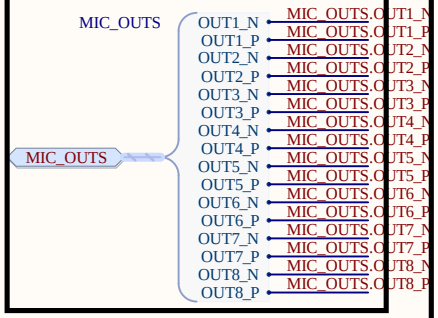
Fixed 3.3V output
150mV dropout @500mA
Max output current is 1A
Enable High to activitate the chip



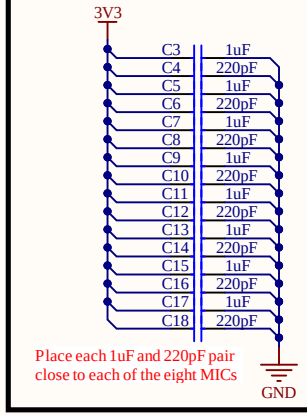
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Title: Power_Section.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 12/30/2022	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 11

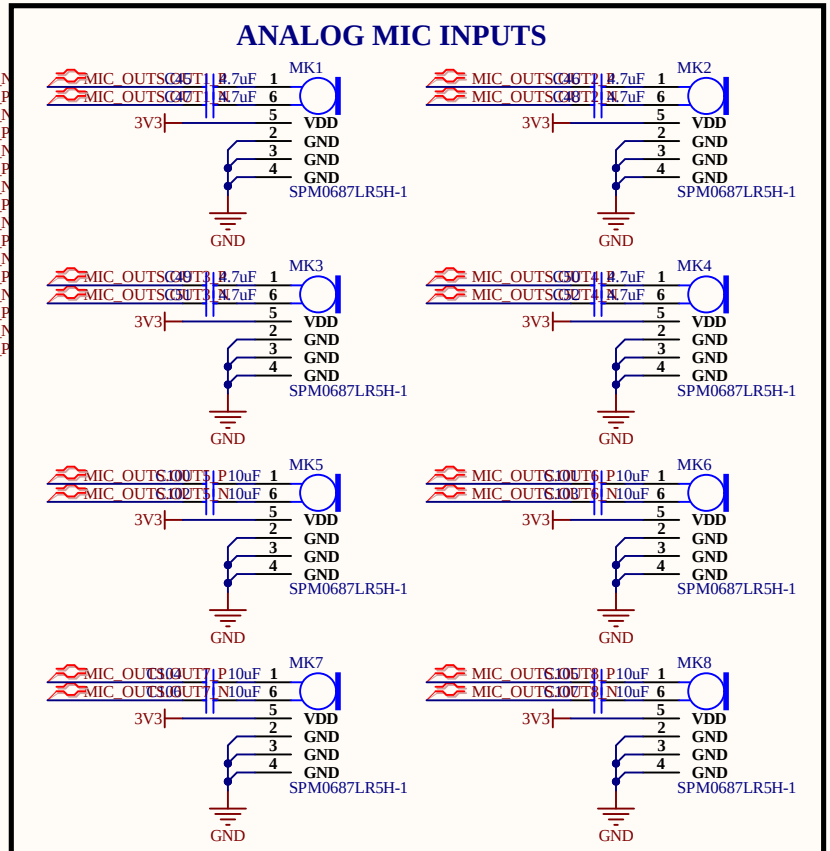
MIC OUT SIGNALS



DECOUPLING CAPS



ANALOG MIC INPUTS

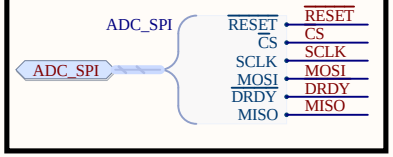




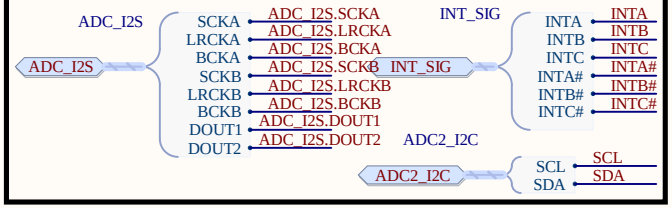
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Sensor_Sheet.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 12/27/2022	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 11

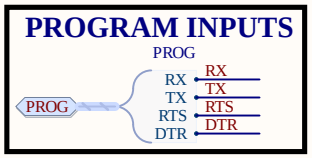
ADC1 OUTPUT



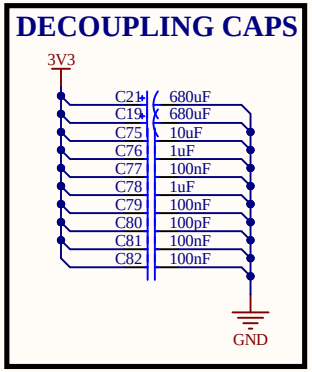
ADC 2 OUTPUT SIGNALS



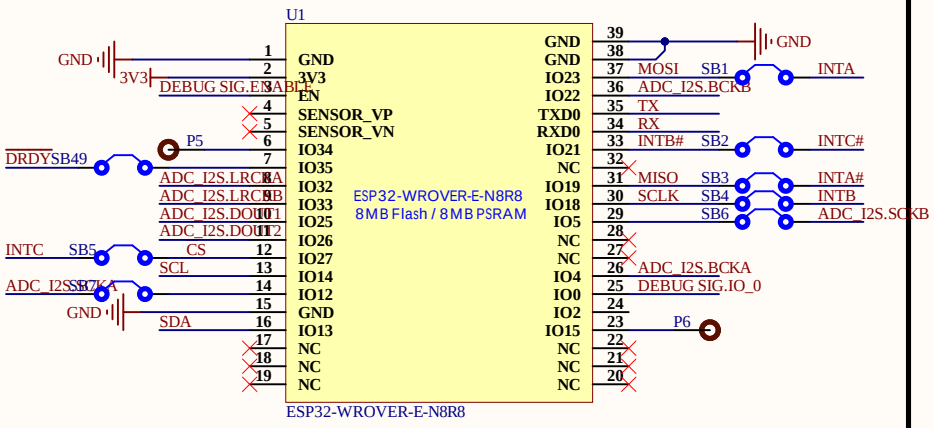
PROGRAM INPUTS



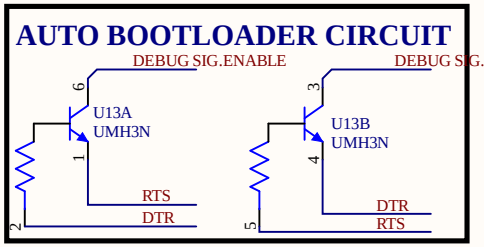
DECOUPLING CAPS



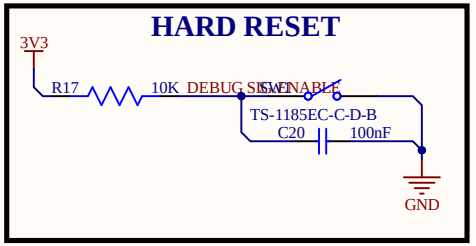
ESP32 Module



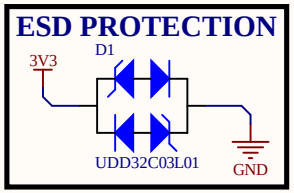
AUTO BOOTLOADER CIRCUIT

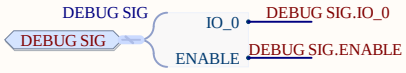


HARD RESET



ESD PROTECTION



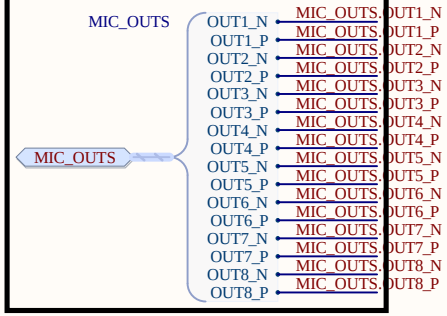




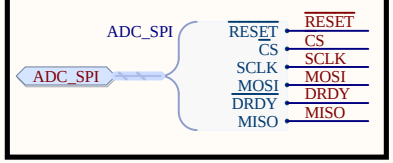
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Title: Controller_Sheet.SchDoc		
File: Sound_Sensor.PrjPcb		
Date: 12/27/2022	Page	Num: 4
Drawn By: Huzaifa Ali	Page	Total: 11

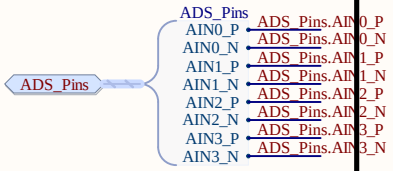
MIC OUT SIGNALS



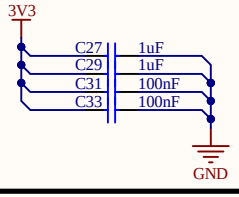
ADC OUTPUT



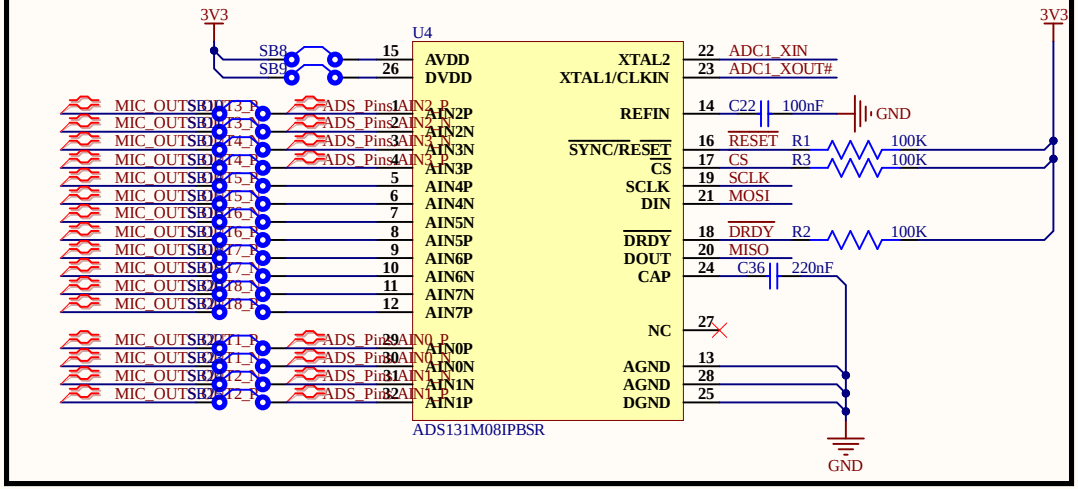
ADS131 SIGNALS



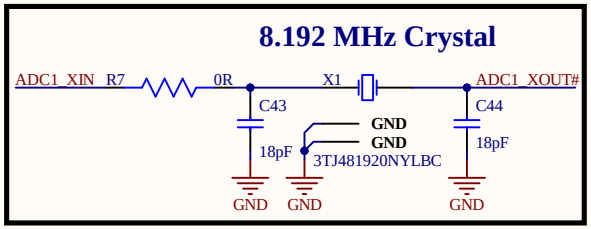
DECOUPLING CAPS



ADC WITH SPI INTERFACE



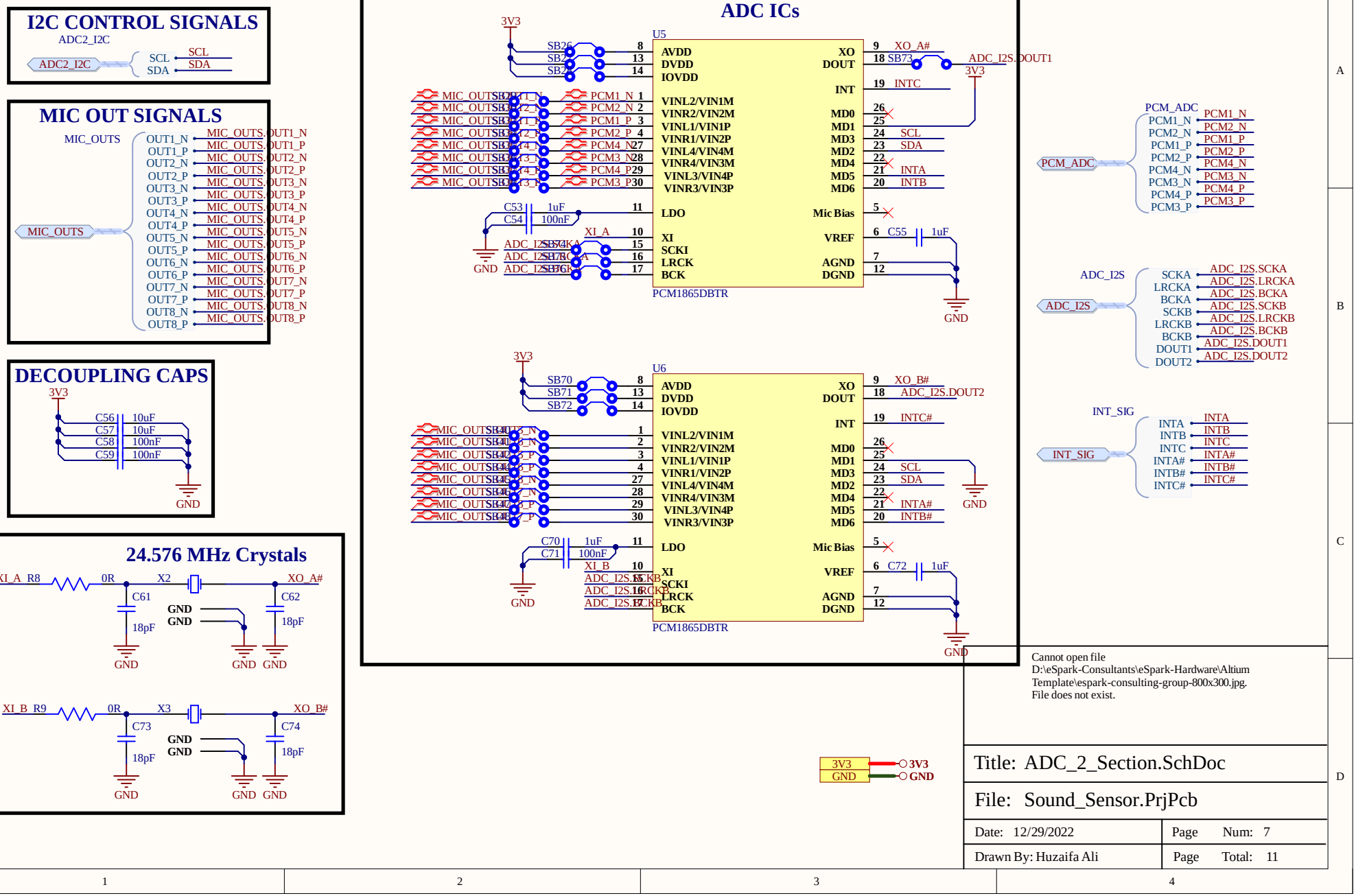
8.192 MHz Crystal



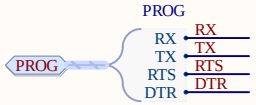


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Title: ADC_1_Section.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 12/29/2022	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 11


I2C CONTROL SIGNALS
ADC2_I2C
ADC2_I2C
SCL
SDA
MIC OUT SIGNALS
MIC_OUTS
OUT1_N
OUT1_P
OUT2_N
OUT2_P
OUT3_N
OUT3_P
OUT4_N
OUT4_P
OUT5_N
OUT5_P
OUT6_N
OUT6_P
OUT7_N
OUT7_P
OUT8_N
OUT8_P
DECOUPLING CAPS
3V3
C56 10uF
C57 10uF
C58 100nF
C59 100nF
GND
24.576 MHz Crystals
XI_A_R8
0R
X2
XO_A#
C61
C62
18pF
18pF
GND
XI_B_R9
X3
XO_B#
C73
C74
ADC ICs
U5
AVDD
DVDD
IOVDD
VINL2/VIN1M
VINR2/VIN2M
VINL1/VIN1P
VINR1/VIN2P
VINL4/VIN4M
VINR4/VIN3M
VINL3/VIN4P
VINR3/VIN3P
LDO
XI
SCKI
LRCK
BCK
XO
DOUT
INT
MD0
MD1
MD3
MD2
MD4
MD5
MD6
Mic Bias
VREF
AGND
DGND
PCM1865DBTR
C53 1uF
C54 100nF
C55 1uF
XI_A
XO_A#
INTC
SCL
SDA
INTA
INTB
U6
XI_B
XO_B#
INTC#
INTA#
INTB#
C70 1uF
C71 100nF
C72 1uF
PCM_ADC
ADC_I2S
INT_SIG
SCKA
LRCKA
BCKA
SCKB
LRCKB
BCKB
DOUT1
DOUT2
ADC_I2S.SCKA
ADC_I2S.LRCKA
ADC_I2S.BCKA
ADC_I2S.SCKB
ADC_I2S.LRCKB
ADC_I2S.BCKB
ADC_I2S.DOUT1
ADC_I2S.DOUT2
INTA
INTB
INTC
INTA#
INTB#
INTC#
3V3
GND
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Template\espark-consulting-group-800x300.jpg.
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Title: ADC_2_Section.SchDoc
File: Sound_Sensor.PrjPcb
Date: 12/29/2022
Page Num: 7
Drawn By: Huzaifa Ali
Page Total: 11

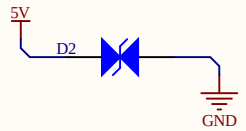
PROGRAM INPUTS



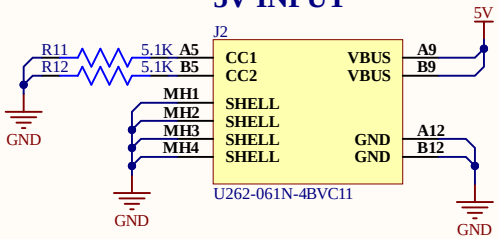
DEBUGGING SIGNALS



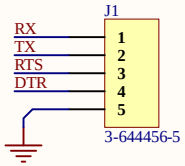
ESD PROTECTION



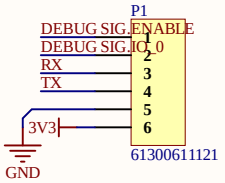
5V INPUT



PROGRAMMING HEADER



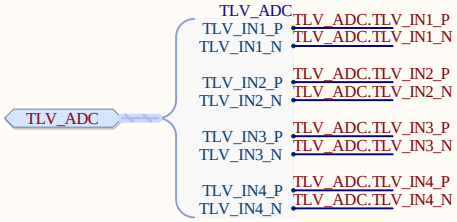
DEBUG CONNECTOR

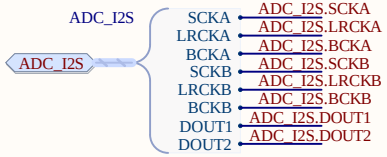


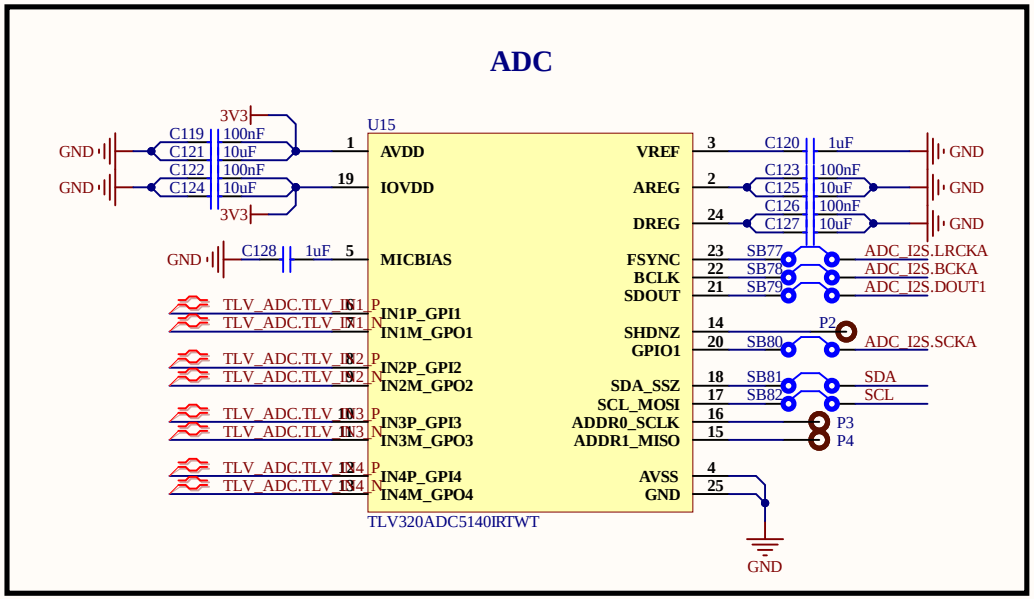


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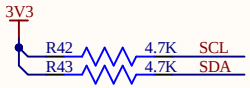
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File: Sound_Sensor.PrjPcb	
Date: 12/29/2022	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 11







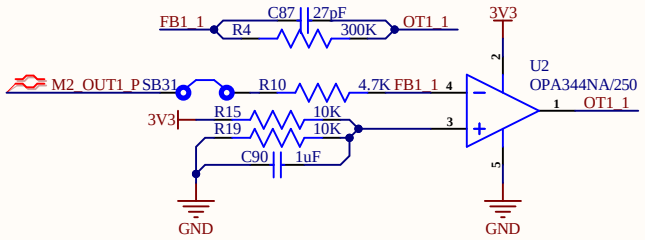


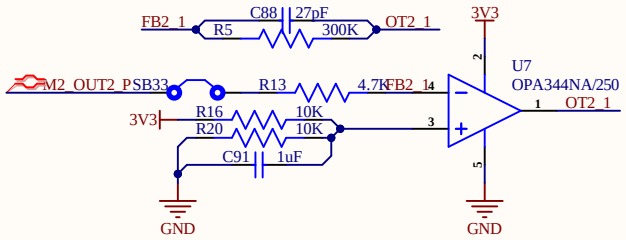


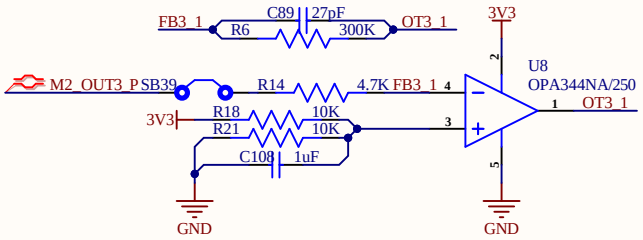


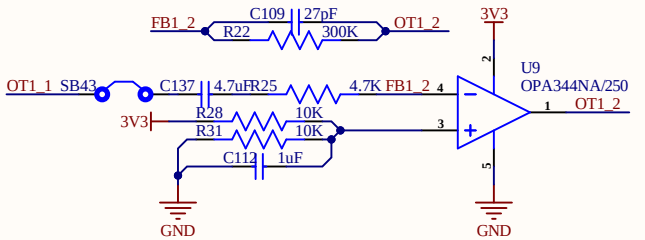
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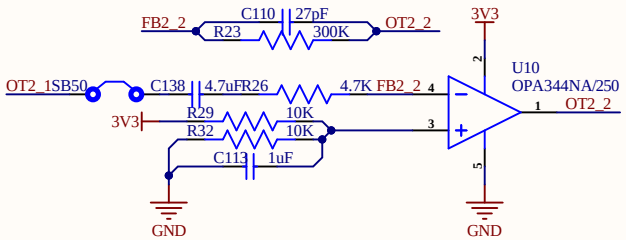
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Date: 12/27/2022	Page Num: *
Drawn By: Huzaifa Ali	Page Total: 11

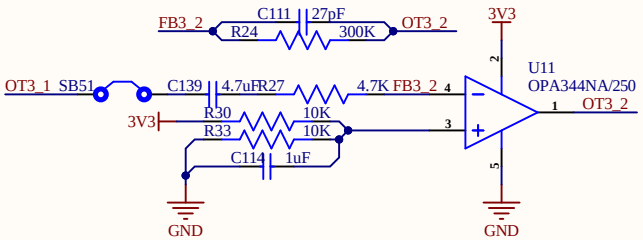


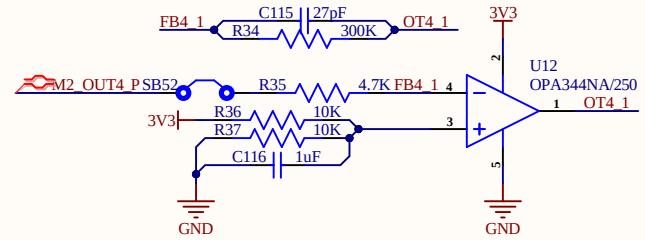


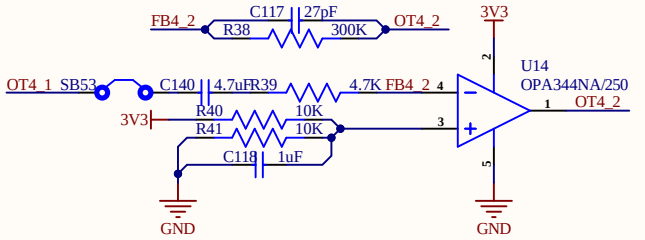


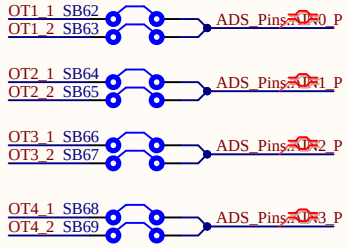


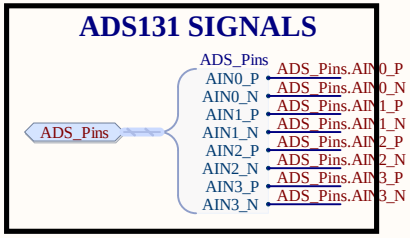




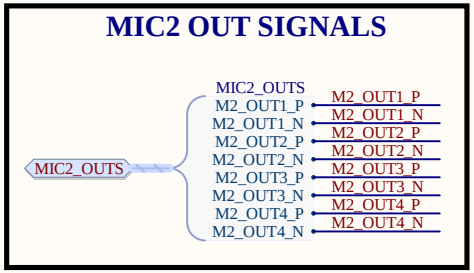






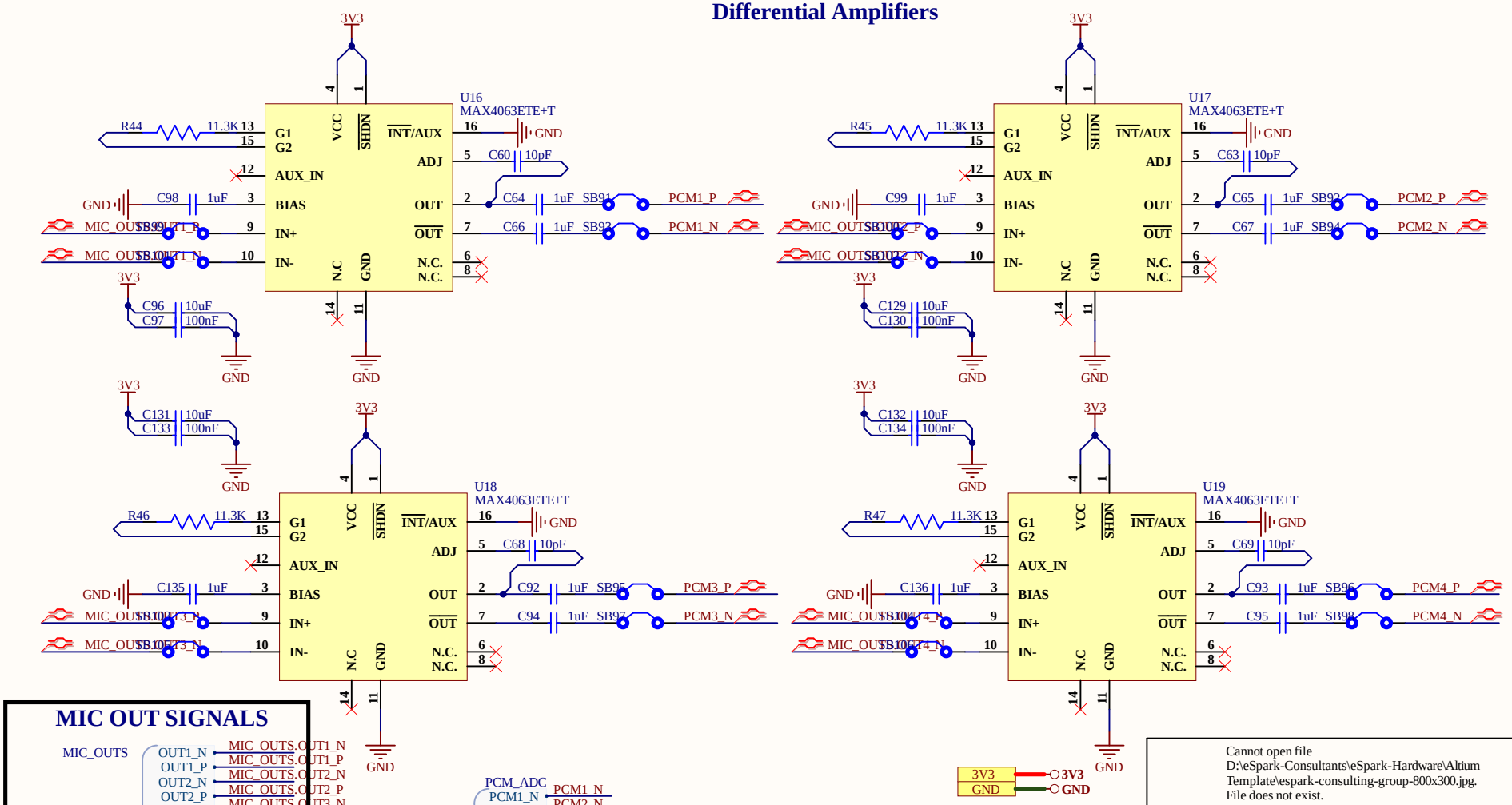




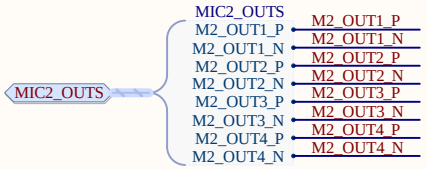


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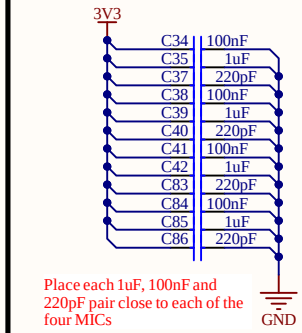
Title: Amplifier_Sheet.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 12/30/2022	Page Num: *
Drawn By: Huzaifa Ali	Page Total: 11


Differential Amplifiers
U16
MAX4063ETE+T
U17
MAX4063ETE+T
U18
MAX4063ETE+T
U19
MAX4063ETE+T
3V3
GND
R44
11.3K
R45
R46
R47
C98
C99
C135
C136
C96
C97
C129
C130
C131
C133
C132
C134
C60
C63
C68
C69
C64
C65
C92
C93
C66
C67
C94
C95
PCM1_P
PCM1_N
PCM2_P
PCM2_N
PCM3_P
PCM3_N
PCM4_P
PCM4_N
G1
G2
VCC
SHDN
INT/AUX
ADJ
AUX_IN
BIAS
OUT
IN+
IN-
N.C.
MIC OUT SIGNALS
MIC_OUTS
PCM_ADC
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D:\eSpark-Consultants\eSpark-Hardware\Altium
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Date: 12/29/2022
Page Num: *
Drawn By: Huzaifa Ali
Page Total: 11

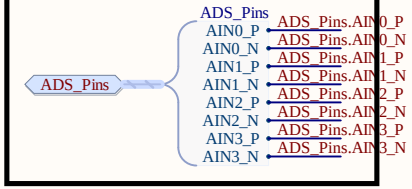
MIC2 OUT SIGNALS



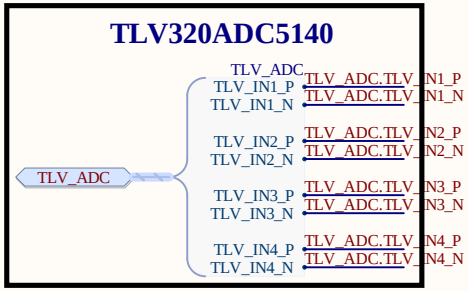
DECOUPLING CAPS



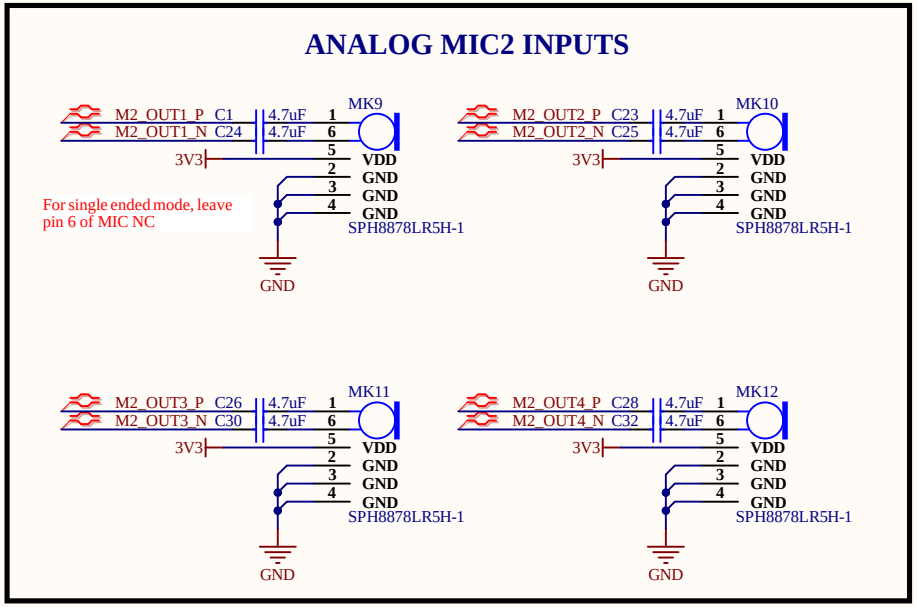
ADS131 SIGNALS



TLV320ADC5140



ANALOG MIC2 INPUTS



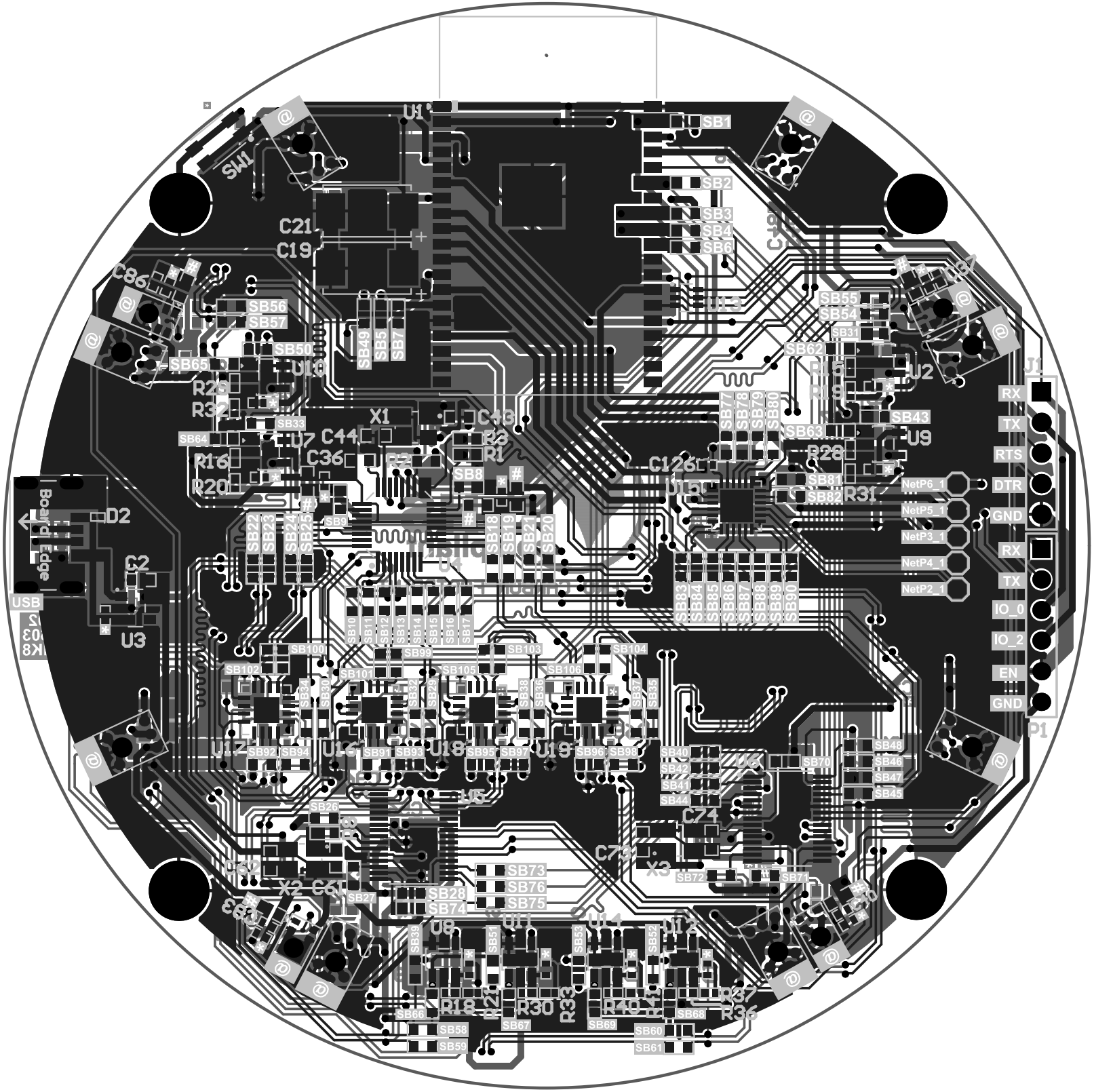
Selection





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Title: Sensor_Sheet_2.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 12/29/2022	Page Num: *
Drawn By: Huzaifa Ali	Page Total: 11


U1
SW1
C21
C19
C86
SB56
SB57
SB50
SB65
U10
R26
R32
SB33
SB64
U7
C44
X1
C43
C36
R16
R20
R3
R1
SB8
SB9
SB22
SB23
SB24
SB25
SB18
SB19
SB21
SB20
D2
C2
U3
USB
Board Edge
SB100
SB102
SB101
SB99
SB103
SB104
SB105
SB106
U12
U13
U14
U15
U16
U17
U18
U19
SB92
SB94
SB91
SB93
SB95
SB97
SB96
SB98
U5
SB26
SB27
X2
C61
C62
C63
SB73
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SB77
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SB80
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U15
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R31
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NetP3_1
NetP4_1
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SB90
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RTS
DTR
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RX
TX
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IO_2
EN
GND
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SB42
SB41
SB44
SB70
SB48
SB46
SB47
SB45
C24
C73
X3
SB72
SB71
C10